Exact conditional module tests with external dependence structure for replicable response biology in small immunotherapy cohorts

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Abstract

Public immune-checkpoint-inhibitor (ICI) cohorts provide tumour transcriptomes for only 25–300 patients each. That is too few to estimate the gene–gene dependence that conditional inference requires, and response is often confounded with batch structure. Consequently, transcriptomic predictors of response rarely replicate. We present a framework that learns dependence structure (a sparse gene graph and latent-factor directions) once from large unlabelled external data, and eliminates all cohort-specific parameters by conditioning rather than estimation. The resulting gene-module test is exact in finite samples for any outcome distribution whenever the external structure contains the cohort's true structure. Evidence is combined across cohorts by a direction-consistent partial-conjunction test that is valid whenever the cohort-level tests are exact, and across modules by false discovery rate (FDR) procedures. We show that misspecified structure induces an omitted-variable bias whose effect on FDR grows with sample size, and that over-spanning the external factor space removes it when the error lies in the factor span, though not when response-related dependence is missing from the graph. In simulations the test is calibrated in every setting and costs little power relative to an oracle, whereas marginal module tests and label-aware batch correction inflate FDR up to 2.8- and 4.5-fold. Under adversarial factor-span misspecification, over-spanning restores FDR to 0.058–0.112. In a pre-registered, hash-sealed analysis of 1009 patients across ten ICI cohorts, no tested module carried detectable response information beyond the established T-effector/IFN-γ signature, and discovery p-values were consistent with the global null. These results provide a leakage-free procedure for conditional biomarker discovery in small cohorts.

Keywords: conditional independence testing; false discovery rate control; immune checkpoint inhibitors; transcriptomic biomarkers; replicability; partial conjunction testing; external data integration; Gaussian graphical models; batch effects; pre-registration.

Code and data: `https://github.com/encoder7-b/RECOMB-PAPER` (code, simulations, protocol, label hashes and fetch scripts for all public data).

1 Introduction

Many studies have proposed transcriptomic predictors of response to immune checkpoint blockade. Few hold up in independent cohorts. Three features of the public data explain much of this.

- **Small cohorts.** Typical cohorts have 25–300 patients and 20,000 genes (Hugo et al., 2016; Riaz et al., 2017; Gide et al., 2019; Liu et al., 2019; Jung et al., 2019; Mariathasan et al., 2018; McDermott et al., 2018; Braun et al., 2020).
- **Response is entangled with batch:** drug, regimen, prior therapy, biopsy site and trial.
- **Leakage.** Cross-cohort correction that protects the outcome as a covariate inflates downstream significance (Nygaard et al., 2016).

We ask a sharper question than "is module G associated with response?" We ask whether G carries information *beyond all other measured genes*, including the established T-effector/IFN-γ signature (Ayers et al., 2017), and whether that holds in more than one cohort.

Related work. Table 1 compares the closest methods. Model-X knockoffs and the conditional randomisation test (CRT) control error for conditional associations given the feature distribution (Candès et al., 2018). The multi-environment knockoff filter adds cross-environment consistency (Li et al., 2022). Conditional knockoffs condition on sufficient statistics to avoid estimating parameters (Huang and Janson, 2020), and summary-statistic knockoffs borrow dependence from reference panels (He et al., 2022). A recent CRT variant uses external and unlabelled data to estimate the conditional distribution by transfer learning and a residual bootstrap (CRT* authors, 2026). Our method differs in what it takes from external data (*structure only*: a graph support and a factor span). It also differs in how it handles everything else (exact elimination by conditioning, in closed form), and in its unit and target (gene modules, replicated across independent small cohorts with batch). Table 1 compares assumptions and guarantees, not performance. The only empirical comparison with a knockoff-based multi-environment pipeline is in its oracle form (our first design, below), and we did not run CRT*.

A negative result motivated the design. Our first design combined conditional knockoffs on an external graph with derandomised e-values (Ren and Barber, 2024) and partial conjunction. It made *zero* discoveries in all 288 oracle settings we ran (knockoffs from the true model, n up to 300, AUC up to 0.92), because knockoff+ needs more clean selections than 10–20 true signals can supply. Per-hypothesis tests do not have this floor.

Contributions.

1. An exact conditional test for a gene set. External data supply only a graph support and a factor span, and every cohort-specific parameter is eliminated by conditioning (Proposition 1, full proof in Appendix A).
2. A direction-consistent Stouffer partial-conjunction combiner, with multiplicity control across modules (BH, and BY under arbitrary dependence).
3. An omitted-variable characterisation of misspecification: inflation grows like $\sqrt{n}$ and is driven by the residual correlation between the omitted direction and the outcome residual (Proposition 2). Simulations confirm this. Over-spanning repairs factor-span errors, but it cannot repair omitted response-related graph edges, which inflate FDR up to 0.38.
4. Evidence that label-aware batch correction and marginal module tests do not control FDR for this estimand.
5. A pre-registered, hash-sealed analysis of 1009 patients, with a sealed replication audit, a real-data permutation null and exploratory checks.

2 Methods

2.1 Model and estimand

In cohort k, row i holds expression $X_i \in \mathbb{R}^p$, a response Y_i and a batch label b_i from a finite set. A marker set B is never tested. For a gene set G disjoint from B, the null is

$$H_{G,k}: \; Y_i \perp\!\!\!\perp X_{iG} \mid X_{i,-G}, b_i.$$

Assumption 1 (Sampling)**.** *Given the batch labels, the rows (X_i, Y_i) are independent and identically distributed within each batch.*

Assumption 2 (Factor-plus-graph Gaussian model)**.** $X_i \mid b_i \sim N(\mu_{b_i}, \Sigma)$ *with* $\Sigma = \Lambda\Lambda^\top + \Omega^{-1}$*, where* $\Lambda \in \mathbb{R}^{p\times d}$*,* $\Omega \succ 0$*, and* $\Omega_{jl} \neq 0$ *only on the edges of a graph* $\mathcal{E}$*. Batch shifts only the mean.*

Table 1: Comparison with the closest methods.

Method	What must be known or estimated about X	Use of external data	Finite-sample guarantee	Unit	Multi-cohort replicability
Model-X knockoffs / CRT (Candès et al., 2018)	full $P(X)$	none (or to estimate $P(X)$)	exact if $P(X)$ known	feature	no
Multi-environment knockoffs (Li et al., 2022)	$P(X)$ per environment	none	exact if each $P(X)$ known	feature	yes (partial conjunction)
Conditional knockoffs (Huang and Janson, 2020)	model family (e.g. graph)	none	exact in model; needs n large vs. local structure	feature	no
GhostKnockoff (He et al., 2022)	LD from reference panel	reference covariance	approximate	variant	no
CRT* (CRT* authors, 2026)	$X \mid Z$ estimated (transfer learning)	estimates the conditional law	approximate (bootstrap)	single X	no
This work	graph support and factor span only	structure only; parameters conditioned away	**exact** under containment (Prop. 1)	module or gene	yes (Lemma 1)

Assumption 3 (Containment)**.** *The external structure $(\hat{\mathcal{E}}, \hat{W})$ is independent of the cohort data and satisfies two conditions.*

- *$N(G) \subseteq \hat{N}(G)$, where $N(G)$ is the set of genes outside G that are adjacent to G in $\mathcal{E}$.*
- $\mathrm{colspan}(\Omega\Lambda) \subseteq \mathrm{colspan}(\hat{W})$.

The design for G is $D_G = [\mathbf{1},\ \text{batch dummies},\ X_{\hat{N}(G)},\ X_{-G}\hat{W}_{-G}]$, with rank q. The residual η is any function of (Y, X_{-G}, b). We use Y minus its least-squares fit on batch and the mean of B. Nothing used to build D_G, η, a projection of X_G, or a rule for skipping a test may depend on X_G.

2.2 The exact test

Let $\tilde{X}_G$ and $\tilde{\eta}$ be the residuals of X_G and η after projection onto the columns of D_G. Define

$$u = \tilde{X}_G^\top \tilde{\eta}/\|\tilde{\eta}\|, \qquad S = \tilde{X}_G^\top \tilde{X}_G - uu^\top.$$

Wilks' statistic for adding η to the regression of X_G on D_G is $1/(1+u^\top S^{-1}u)$. The test is undefined when $\tilde{\eta} = 0$ (for example, constant Y); that event is a function of the conditioning variables, so skipping it is harmless.

Proposition 1 (Exactness)**.** *Suppose Assumptions 1–3 and $H_{G,k}$ hold, $g = |G|$, $\tilde{\eta} \neq 0$ and $n - q - 1 \geq g$. Then, conditionally on (X_{-G}, Y, b),*

$$F = \frac{n-q-g}{g}\, u^\top S^{-1} u \ \sim\ F_{g,\, n-q-g}.$$

Because $q = \mathrm{rank}(D_G)$ is itself a function of the conditioning variables, the p-value $\bar{F}_{g,n-q-g}(F)$ is exactly uniform, conditionally and therefore unconditionally. This holds for every conditional law of Y, and it does not depend on μ, Σ or the batch effects. For $g = 1$, or for any fixed projection $X_G a$, it reduces to an exact t-test with $n - q - 1$ degrees of freedom, whose sign is used.

The proof is in Appendix A. Its three steps:

1. By the Woodbury identity, $E[X_G \mid X_{-G}, b]$ is linear in the neighbours $X_{N(G)}$ and in the d combinations $X_{-G}(\Omega\Lambda)_{-G}$, with constant conditional covariance. Under containment this is $D_G\beta$.
2. Under $H_{G,k}$ with independent rows, conditioning additionally on Y leaves this law unchanged, so D_G and η can be treated as fixed.
3. With a single added regressor, $u \sim N_g(0, \Sigma_c)$ is independent of $S \sim \mathrm{Wishart}_g(n - q - 1, \Sigma_c)$. So $(n - q - 1)u^\top S^{-1}u$ is Hotelling's T^2, which is exactly F-distributed for any g.

Only the conditional law of X_G given (X_{-G}, b) is used, so sampling that depends on Y (for example, case–control selection by response) does not affect validity. Genes are scaled by fixed external standard deviations D. This changes nothing: $\Omega' = D\Omega D$ has the same support and $W' = DW$, so containment is simply read in the scaled coordinates. Scaling by cohort standard deviations would break exactness, because they depend on X_G. Containment is sufficient but not necessary: the proof needs only the g columns of $W_{-G}MW_G^\top$ (Appendix A) to lie in $\mathrm{colspan}(\hat{W}_{-G})$.

Table 2: Exactness under the conditional null: 20,000 replicates per row, 95% Monte Carlo intervals.

n	g	reject @0.05	reject @0.01	KS p
15	1	0.0520 ± 0.0030	0.0097 ± 0.0014	0.75
25	1	0.0500 ± 0.0030	0.0086 ± 0.0014	0.39
25	5	0.0503 ± 0.0030	0.0112 ± 0.0014	0.29
50	10	0.0505 ± 0.0030	0.0101 ± 0.0014	0.55
100	10	0.0505 ± 0.0030	0.0114 ± 0.0014	0.18

Verification. Our implementation agrees with independent implementations (statsmodels OLS for $g = 1$, and multivariate OLS/MANOVA Wilks for $g = 5$) to within 8e-14 in $|\log p|$. A numerical audit of Lemma A1 over 200 random sparse-plus-factor models confirms that the true conditional-mean coefficients lie in the span of the true neighbours and $\Omega\Lambda$, and in any span containing $\Omega\Lambda$, to within 10^{-13}. Dropping a single neighbour or factor direction breaks this. Under the conditional null, with a binary Y that depends nonlinearly on genes outside D_G and 20,000 replicates per setting, rejection rates match nominal levels for n from 15 to 100 and g from 1 to 10 (Table 2).

2.3 External structure

The external structure is learned from unlabelled expression of the matched tumour types (TCGA; Cancer Genome Atlas Research Network et al., 2013):

- standardise each gene;
- estimate d principal factors $\hat{\Lambda}$;
- fit a neighbourhood-lasso graph to the factor residuals (Meinshausen and Bühlmann, 2006), take its union with each gene's top correlation partners, and cap the degree at 3;
- estimate $\hat{\Omega}$ on that graph.

Assumption 3 needs containment, not recovery, so we inflate both parts. For the graph we take the union with correlation partners. For the factor span we use $\hat{W} = [\hat{\Omega}\hat{\Lambda},\ \hat{\Lambda}]$ with over-specified $d = \mathrm{clip}(\lfloor n/25 \rfloor, 2, 8)$ (*over-spanning*). Each column costs one degree of freedom, and a module is skipped in a cohort if fewer than 8 remain.

Remark 1. Over-spanning makes containment of the factor span more likely, but it does not guarantee it. It cannot repair an omitted graph edge. Section 3 separates these cases.

2.4 Combining cohorts and modules

Lemma 1 (Partial conjunction)**.** *Suppose the cohorts are independent given the external structure, that in each cohort the decision to test G is a function of (X_{-G}, Y, b), and let $H_G^{r/K}$ state that G is non-null in fewer than r of the K cohorts in which G is tested. For each sign $s \in \{\pm 1\}$, let $p_{(1)}^{(s)} \le \cdots \le p_{(K)}^{(s)}$ be the ordered exact one-sided p-values $1 - \Phi(s z_k)$, and let $m = K - r + 1$. Then*

$$P_G^{r/K} = \min\Big\{1,\ 2 \min_s \bar{\Phi}\Big(\textstyle\sum_{i \ge r} \Phi^{-1}\big(1 - p_{(i)}^{(s)}\big)/\sqrt{m}\Big)\Big\}$$

satisfies $\Pr(P_G^{r/K} \le t) \le t$ under $H_G^{r/K}$, conditionally on the set of tested cohorts. If $K < r$, we set $P_G^{r/K} = 1$.

The idea is that at least m cohorts are null and each z_k is exactly $N(0,1)$ there. The m largest p-values dominate a sorted set of m independent uniforms, and the Stouffer combination is monotone (Appendix A; Benjamini and Heller, 2008).

Across modules the pre-registered procedure is BH (Benjamini and Hochberg, 1995). BH is guaranteed only when the null p-values are positively dependent (PRDS). No PRDS argument is available here: modules share the residual η, their designs overlap, one module's genes sit inside another's X_{-G}, and the order statistic and sign correction are layered on top. BH therefore has no FDR guarantee in this setting. Benjamini–Yekutieli (BY) is valid under arbitrary dependence, and we report it alongside. BY rejects a subset of what BH rejects. Cohort independence is a correctness requirement, which we enforce with a calibrated duplicate-patient screen (Section 4).

Untested cohorts. A module may be untestable in a cohort because too few degrees of freedom remain. This depends only on q, a function of X_{-G}, so dropping that cohort from the module's combination keeps Lemma 1 valid. The implementation has two further skip rules: a module is skipped if most of its genes are constant in the cohort, and the test returns no value if S is numerically singular. Both use X_G, so the lemma as stated does not cover them. Under Assumption 2 both events have probability zero, so they do not affect validity under the model. The pre-registered rule set an untested cohort's z to 0. For $r = 1$, the primary analysis, this is provably conservative: under the global null the combined statistic is $N(0, K_t/K)$, where K_t is the number of tested cohorts. For $r \ge 2$ it is unproven, though conservative in our simulation checks. We report both rules, and they agree (Section 4).

2.5 Misspecification

Proposition 2 (Omitted-direction bias; informal)**.** *Suppose* $E[X_G \mid X_{-G}, b] = D_G\beta + U\gamma$*, where* $U = X_{-G}c \notin \mathrm{colspan}(D_G)$*. This happens when a true edge is missing from* $\hat{\mathcal{E}}$ *or when Assumption 3 fails for* $\hat{W}$*. Take* $g = 1$*, with* γ *and* q *fixed as* n *grows, and assume the sample moments converge. Write* $\kappa = \mathrm{Cov}(\eta, U \mid D_G)/\mathrm{sd}(\eta \mid D_G)$ *and* $\tau^2 = \mathrm{Var}(U \mid D_G, \eta)$*. Then under* $H_{G,k}$ *the t-statistic for* η *satisfies* $t = \delta_n + O_p(1)$*, with*

$$\delta_n = \frac{\sqrt{n}\,\gamma\,\kappa}{\sqrt{\sigma_c^2 + \gamma^2\tau^2}}.$$

The shift is non-zero only when the omitted direction is correlated with the outcome *residual* η ($\kappa \neq 0$). Being correlated with response is not enough. When $\kappa \neq 0$, the type-I error tends to 1 as n grows. In a check with $\gamma = \sigma_c = 1$, $\mathrm{corr}(\eta, U) = 0.3$ and $n = 400$, the mean simulated t was 4.31, against $\delta_n = 4.34$. The external structure is shared by every cohort of a tumour type, so an omitted edge biases those cohorts in the same direction, and replication across cohorts does not protect against it. The derivation is in Appendix A.

Scope of the guarantee. Proposition 1 is exact only under Assumptions 1–3, and outside them we claim no guarantee.

- An omitted direction correlated with η drives the type-I error to 1 (Proposition 2). In adversarial simulations, missing response-related edges inflate FDR up to 0.38.
- Over-spanning repairs factor-span errors only.
- Patients shared between cohorts violate Lemma 1 (Section 3).
- If batch changes variances or loadings rather than only the mean, the conditional coefficients differ by batch and exactness fails.

2.6 Pre-registration

All label-free choices were frozen in a protocol and committed before any outcome was read:

- gene universe (1500 genes) and marker set;
- modules (TCGA residual-correlation clusters);
- batch variables and response definitions;
- sealed cohorts, primary analysis ($r = 1$, BH at 0.1) and replication rule.

Table 3: Global null with external structure ($N_{\text{ext}} = 500$): 2000 replicates per row, 95% intervals.

n	d	FDR, r=1	FDR, r=2	feasible tests: reject @0.05
25	2	0.055 [0.045, 0.065]	0.003 [0.001, 0.005]	0.0502 (52% feasible)
50	2	0.103 [0.090, 0.117]	0.009 [0.004, 0.013]	0.0488 (100% feasible)
100	4	0.096 [0.083, 0.108]	0.004 [0.001, 0.006]	0.0501
300	8	0.097 [0.084, 0.109]	0.001 [0.000, 0.002]	0.0499

The commit includes SHA-256 hashes of every label file, and the analysis refuses to read true labels unless the protocol is committed and every hash matches.

3 Simulations

Design. Each simulated dataset has:

- 520 genes: 50 tested modules of 10, plus 20 marker genes whose mean drives response;
- AR(1) dependence within modules, and four latent factors $F_1, \ldots, F_4$ (F_1 immune, F_4 diffuse);
- a two-level batch with gene-specific shifts and a response-rate shift;
- binary Y with heterogeneous effects across cohorts.

Signal is *sparse* (2 genes per module, random signs) or *coherent* (all genes in a module, one sign). Effect sizes are set by the AUC of the signal genes' combined score, 0.65–0.80. External structure is learned from $N_{\text{ext}} = 500$ or 2000 unlabelled samples. All results use $K = 6$ cohorts and BH at $\alpha = 0.1$, with $r = 1$ unless stated.

Calibration. Table 3 reports 2000-replicate global-null runs with external structure learned from $N_{\text{ext}} = 500$, the size of a TCGA tumour type.

- FDR, which under the global null equals the chance of any false discovery, stays within Monte Carlo error of $\alpha = 0.1$ or below at every n.
- Among feasible tests the per-test rejection rate matches 0.05.
- At $n = 25$ only about half of the modules are feasible. That is a loss of power, not of validity.

Power. With oracle structure and coherent signal at AUC 0.75, power is 0.51 at $n = 100$ and 0.99 at $n = 300$ (Fig. 1). Learning the structure externally retains most of this: 0.37–0.50 at $n = 100$ and 0.96–0.98 at $n = 300$. Replication ($r = 2$) is far costlier below $n \approx 300$, and single-gene tests have almost no power, so modules are the inferential unit.

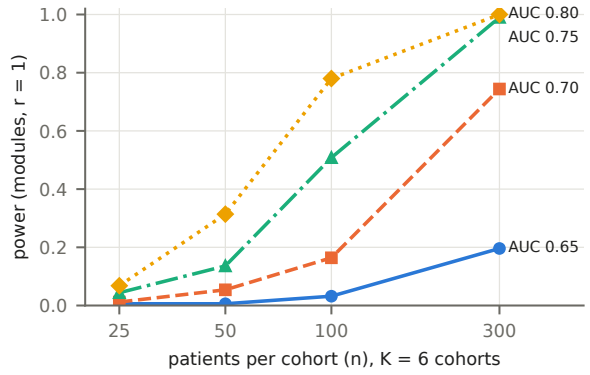


Figure 1: Power with oracle structure (coherent signal, $K = 6$, $r = 1$).

Table 4: FDR and power on the same simulated cohorts (50 replicates). [a]Mean over $n \in \{50, 100, 300\}$ with no signal.

Method	FDR (coherent, AUC 0.75) n=50	n=100	n=300	FDR global null[a]	Power n=300
Ours (external structure)	0.087	0.071	0.086	0.120	0.96
In-sample structure	0.035	0.109	0.089	0.107	0.94
Marginal per cohort	0.154	0.164	0.281	0.100	0.88
Pooled, label-aware correction	0.268	0.257	0.448	0.207	0.83
Pooled, label-blind correction	0.097	0.123	0.278	0.073	0.87

Baselines. Table 4 and Fig. 2 compare five methods on cohorts whose response rates differ. Only the conditional tests controlled FDR.

- Marginal tests absorb shared infiltration, so their FDR grows with n.
- Pooling with label-aware correction (Johnson et al., 2007) inflates FDR even under the global null, whereas label-blind pooling does not. This is the leakage mechanism of Nygaard et al. (2016). With signal present, both pooled variants inflate, because neither conditions on the other modules.
- Structure learned in-sample is valid but less powerful at $n = 100$ (0.21 against 0.34).

Factor-span misspecification. With the graph held at the truth, we varied the factor span directly (Table 5). For each omitted factor we also measured $|\rho|$, the correlation between the omitted direction and η given D_G (the driver in Proposition 2). For two independent vectors of length 300 the noise level is about 0.046.

- FDR stayed at or below nominal for the true span and for a span containing it.

Table 5: FDR ($r = 1$, AUC 0.8) by factor span, with the oracle graph (300 replicates, Monte Carlo SE $\leq$0.013). $|\rho|$: measured correlation between the omitted direction and η given D_G, for sparse/coherent signal.

	sparse signal			coherent signal		
Factor span	n=50	n=100	n=300	n=50	n=100	n=300
true span	0.072	0.076	0.080	0.078	0.078	0.085
true span + 4 random	0.071	0.090	0.080	0.091	0.083	0.071
omit F_1 (immune), $\|\rho\|$=0.027/0.020	0.083	0.065	0.072	0.070	0.061	0.067
omit F_2, $\|\rho\|$=0.052/0.048	0.088	0.086	0.076	0.070	0.079	0.085
omit F_4 (diffuse), $\|\rho\|$=0.057/0.116	0.096	0.072	0.086	0.094	0.106	0.150

- It also stayed nominal when we omitted F_1 (immune) or F_2, whose $|\rho|$ was at or below the noise level. F_1 is strongly linked to response, but that link is absorbed by the marker adjustment in r.
- Omitting the diffuse factor F_4 under coherent signal gave $|\rho|$ of 2.5 times the noise level. FDR rose with n, to 0.150 at $n = 300$.

Correlation with response is therefore not what matters. Correlation with the residual is, as Proposition 2 predicts.

Graph misspecification under adversarial edges. We added 40 true edges joining signal genes to genes in null modules (partial correlation 0.25), then deleted a fraction of them from the learned graph (Fig. 3).

- With the default span ($d = 4$, $\hat{W} = \hat{\Omega}\hat{\Lambda}$), FDR reached 0.20 at $n = 300$ even with nothing deleted, because $\hat{W}$ was estimated with error. With oracle structure it was 0.10.
- Over-spanning restored FDR to 0.058–0.112 with nothing deleted. Power was unchanged at $n = 300$ and lower at $n = 100$ (0.12 against 0.20).
- Deleting true response-related edges still inflates FDR, up to 0.38. Over-spanning cannot repair that.
- Random missing edges, which rarely touch response-relevant genes, left calibration unchanged.
- Distilling the outcome residual with a lasso (Liu et al., 2022) did not help.

Dependent cohorts (shared patients). Lemma 1 assumes independent cohorts. To see what happens when that fails, we made each pair of adjacent cohorts share a fraction of patients (identical X and Y, no batch mismatch, oracle structure, 500 replicates; Table 6).

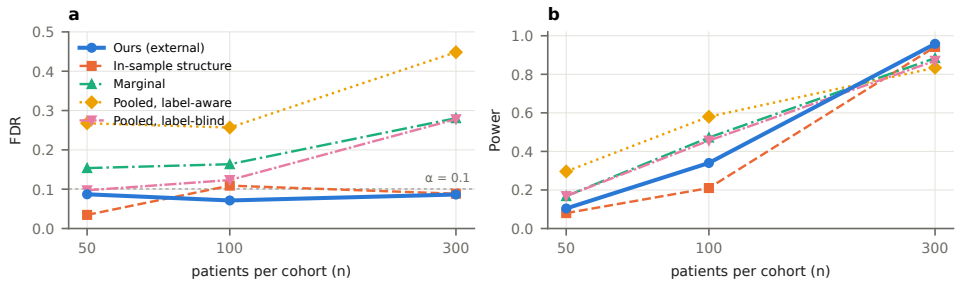


Figure 2: (a) FDR and (b) power by method (coherent signal, AUC 0.75). Dotted line: α.

Table 6: FDR (Monte Carlo SE) with patients shared between adjacent cohorts ($K = 6$, $n = 100$, oracle structure, coherent signal at AUC 0.8, 500 replicates). $r = 1$ unless stated.

Shared patients	FDR, null	FDR, signal	Power, signal	FDR, null, r=2
0%	0.106 (0.014)	0.083 (0.004)	0.79	0.006
10%	0.206 (0.018)	0.117 (0.005)	0.77	0.030
25%	0.366 (0.022)	0.195 (0.006)	0.76	0.044
50%	0.668 (0.021)	0.323 (0.006)	0.73	0.184

- Sharing 10% of patients roughly doubled FDR under the global null at $r = 1$.
- The overlap correlates adjacent z-scores only modestly, but BH's first rejection lies far in the tail, where a slightly wider null distribution matters a great deal.
- Replication at $r = 2$ is more robust.

This is why the duplicate screen is part of the method.

4 Real data

Cohorts. The cohorts are listed in Table 7.

- **Sources:** GEO (Hugo, Riaz, Jung, GSE176307; NCBI GEO, 2021); iAtlas-harmonised cBioPortal studies (Gide, Liu, IMvigor210, IMmotion150; CRI iAtlas, 2024; Cerami et al., 2012); and the Braun supplementary tables.
- **Labels:** response is CR/PR against SD/PD. Jung has no RECIST data in GEO, so we used durable benefit (PFS ≥180 days). Samples are pre-treatment, one per patient.
- **Modules:** the fixed recipe produced 39 modules covering about 400 of the 1500 genes.

Duplicate screen. Each gene is ranked within its cohort, five principal components are removed, and samples are correlated across cohorts. We calibrated this on a positive control: Hugo from GEO against the iAtlas re-processing, the same 26 patients through different pipelines. True pairs scored $\rho \geq 0.67$, non-pairs ≤ 0.27, and impossible NSCLC–ccRCC pairs ≤ 0.28. Ten pairs crossed the threshold of 0.5. None was plausible as a shared patient, and all 14 samples involved were excluded. Assumption 3 also needs the TCGA reference to share no patients with the cohorts, so we ran the same screen of every cohort against every TCGA type. One pair crossed 0.5 (IMmotion150 against TCGA-KIRC, $\rho = 0.50$), well below the true-pair range, and the TCGA sample also correlated at 0.34–0.40 with other IMmotion150 patients, which suggests shared tumour biology. Removing that patient also gave 0 discoveries.

Table 7: Cohorts after duplicate exclusion.

Cohort	Tumour	n	Resp.	Role	Batch
Hugo 2016	melanoma	26	14	discovery	–
Riaz 2017	melanoma	49	10	discovery	–
Gide 2019, PD-1	melanoma	41	19	discovery	–
Gide 2019, ipi+PD-1	melanoma	32	21	discovery	–
Liu 2019	melanoma	121	47	discovery	prior anti-CTLA-4
Jung 2019	NSCLC	27	7	discovery	–
GSE176307	urothelial	87	16	discovery	IO line
IMmotion150, atezo ± bev	ccRCC	160	48	discovery	bevacizumab
IMvigor210	urothelial	298	68	sealed	biopsy site
Braun 2020, nivolumab	ccRCC	168	38	sealed	trial
IMmotion150, sunitinib	ccRCC	81	24	control	–
Braun 2020, everolimus	ccRCC	106	5	control	–

Design. Eight cohorts (543 patients) were used for discovery. IMvigor210 and Braun nivolumab (466 patients) were sealed for replication. The sunitinib and everolimus arms (187 patients) were descriptive controls.

Pre-registered result. There were 0 discoveries at $r = 1$ and $r = 2$, with BH, BY and Simes. The marginal and pooled baselines also made none. Dropping untested cohorts instead of setting $z = 0$ (9 of 312 module–cohort cells were untested) also gave 0 discoveries, with minimum $P = 0.0085$.

- The strongest module, the metallothioneins

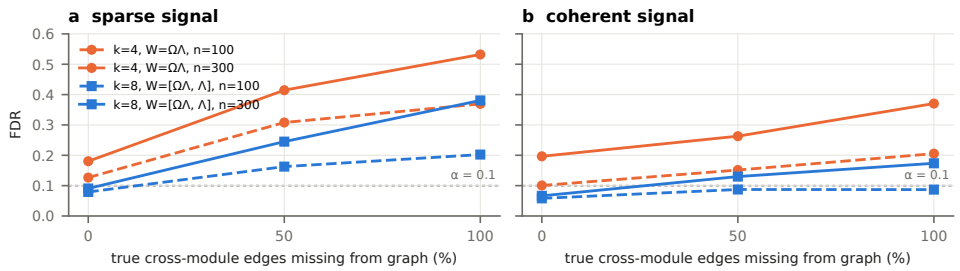


Figure 3: FDR under adversarial missing edges (AUC 0.8). Orange: default span. Blue: over-spanned. Solid $n = 300$, dashed $n = 100$.

MT1G/A/H/M/E, had $P = 0.010$. A single BH rejection would need $P \leq 0.0026$. Its evidence was positive in 7 of 8 discovery cohorts but negative in both sealed cohorts.

- Discovery p-values were consistent with uniform: 2 of 39 fell below 0.05, and the Kolmogorov–Smirnov test against uniform gave $p = 0.13$ (Fig. 4a).
- Sealed-cohort "replication" occurred for 5 modules, against 3.8 expected by chance.

Real-data permutation null. We permuted response within the batch strata of every discovery cohort 1000 times. This keeps the real expression covariance and the real external structure.

- The per-test rejection rate at 0.05 was 0.047.
- The chance of any discovery was 0.084 with BH and 0.021 with BY at $r = 1$, and 0.005 with BH at $r = 2$.
- The real minimum p-value sat at the 0.28 quantile of its permutation distribution.

Exploratory checks (post-unsealing, logged as deviations).

- **T-effector score** (Fig. 4b). It was predictive in Gide, Jung, IMmotion150 and IMvigor210. It was null in both non-ICI controls, and null in Braun and Hugo, as those studies reported (Braun et al., 2020; Hugo et al., 2016). So the labels and pipeline behave as expected.
- **Incremental prediction** (Table 8). Models trained on the discovery cohorts and scored on held-out cohorts gained nothing over the T-effector score. That held when adding any single module (ΔAUC ≤ 0.035) and when adding an elastic net over all 1500 genes (146 non-zero genes).

Table 8: Exploratory held-out AUC [95% bootstrap CI]. Training on discovery cohorts.

Test cohort	T-effector only	+ elastic net (1500 genes)
IMvigor210 (sealed)	0.59 [0.51, 0.67]	0.55 [0.47, 0.62]
Braun nivolumab (sealed)	0.49 [0.39, 0.59]	0.50 [0.40, 0.60]
IMmotion150 sunitinib (control)	0.51 [0.38, 0.64]	0.39 [0.26, 0.52]
Braun everolimus (control)	0.60 [0.27, 0.87]	0.41 [0.23, 0.59]

5 Discussion

The method separates what small cohorts can supply from what they cannot. External data supply the dependence structure. Each cohort's nuisance parameters are eliminated exactly by conditioning. Validity then rests on containment, and the simulations show which violations of containment matter: those whose omitted direction is correlated with the outcome residual. Over-spanning repairs errors that lie in the factor span. It cannot repair omitted response-related edges, which inflated FDR up to 0.38 in our adversarial simulations. Shared patients between cohorts also break the guarantee, which is why the duplicate screen is part of the method rather than a preprocessing detail.

The real-data result is a null for the tested modules. It says that the analysis found no significant incremental conditional association among 39 TCGA-derived modules beyond the T-effector score. It does not show that no other gene set carries such information.

Limitations.

- Exactness assumes conditional Gaussianity on the log scale.
- The modules cover a minority of genes.

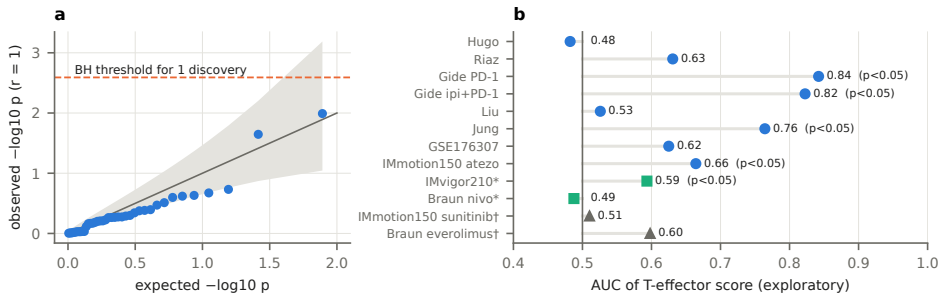


Figure 4: (a) QQ plot of pre-registered discovery p-values with a 95% band. Dashed line: BH threshold for one discovery. (b) Exploratory AUC of the T-effector score. Circles: discovery. Squares (*): sealed. Triangles (†): non-ICI controls.

- Power is low below about 100 patients per cohort.
- BH is not guaranteed under the dependence between modules (BY is).
- The Braun expression matrix was used as released by the authors.
- The estimand is conditional association, not causation or clinical utility.

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A Proofs

Lemma A1 (conditional mean). Write $W = \Omega\Lambda$ and $M = (I_d + \Lambda^\top \Omega \Lambda)^{-1}$. By the Woodbury identity, the precision is

$$\Theta = \Sigma^{-1} = \Omega - WMW^\top.$$

For a fixed batch b, Gaussian conditioning gives

$$E[X_G \mid X_{-G} = x, b] = \mu_{bG} - \Theta_{GG}^{-1}\Theta_{G,-G}(x - \mu_{b,-G}), \qquad \mathrm{Var}(X_G \mid X_{-G}, b) = \Theta_{GG}^{-1} =: \Sigma_c.$$

Now $\Theta_{G,-G} = \Omega_{G,-G} - W_G M W_{-G}^\top$, and the columns of $\Omega_{G,-G}$ are zero outside $N(G)$. So

$$-\Theta_{GG}^{-1}\Theta_{G,-G}x = x_{N(G)}A_1 + (x^\top W_{-G})A_2$$

for some matrices A_1 and A_2. The terms in μ are linear in the batch dummies. Under Assumption 3, $W = \hat{W}C$ for some matrix C. Deleting rows preserves this, so $W_{-G} = \hat{W}_{-G}C$, and $X_{N(G)}$ is a subset of the columns of $X_{\hat{N}(G)}$. Hence $E[X_G \mid X_{-G}, b] = D_G\beta$, with constant covariance Σ_c that does not depend on b. Only the factor term $W_{-G}MW_G^\top\Sigma_c$ of the coefficient matrix uses $\hat{W}$, so it suffices that its g columns lie in $\mathrm{colspan}(\hat{W}_{-G})$.

Lemma A2 (conditioning on Y). By Assumption 1 the rows are independent given b. Under $H_{G,k}$, $Y_i \perp\!\!\!\perp X_{iG} \mid (X_{i,-G}, b_i)$. Therefore the joint conditional law of the rows X_{iG} given (X_{-G}, Y, b) is the product of the laws given $(X_{i,-G}, b_i)$, that is,

$$X_G = D_G\beta + \mathcal{E},$$

where $\mathcal{E}$ has i.i.d. $N(0, \Sigma_c)$ rows independent of (X_{-G}, Y, b). Both D_G and η are functions of (X_{-G}, Y, b), so they are fixed under this conditioning. The residual η may depend on Y in any way. Only Gaussianity of X_G given (X_{-G}, b) was used.

Lemma A3 (exact law for one added regressor). Condition on (X_{-G}, Y, b) and assume $\tilde{\eta} \neq 0$. Let P be the projection onto $\mathrm{colspan}(D_G)$, let $v = \tilde{\eta}/\|\tilde{\eta}\|$ (a unit vector orthogonal to $\mathrm{colspan}(D_G)$), and let $Q = I - P - vv^\top$, a projection of rank $\nu = n - q - 1$.

- Since $(I - P)D_G = 0$, we have $u = \tilde{X}_G^\top v = \mathcal{E}^\top v \sim N_g(0, \Sigma_c)$.
- Similarly $S = \mathcal{E}^\top Q\mathcal{E} \sim \mathrm{Wishart}_g(\nu, \Sigma_c)$.
- u and S are independent, because $Qv = 0$ and the rows of $\mathcal{E}$ are Gaussian.

Hence $T^2 = \nu\, u^\top S^{-1} u$ is Hotelling's $T^2_{g,\nu}$, and $\frac{\nu-g+1}{g\nu}T^2 \sim F_{g,\nu-g+1}$ whenever $\nu \geq g$. Since $\nu - g + 1 = n - q - g$, this proves Proposition 1.

Wilks' statistic is $\det S / \det(S + uu^\top) = 1/(1 + u^\top S^{-1} u)$. The conditional law depends on the conditioning values only through q, so the p-value is exactly uniform conditionally, and hence unconditionally. Replacing X_G by a fixed projection $X_G a$ gives the $g = 1$ case, $t = \sqrt{\nu}\, u/\sqrt{S} \sim t_\nu$, with the sign carried by u. The law involves neither β nor Σ_c, which is the sense in which the nuisance parameters are eliminated.

Proof of Lemma 1. Fix s. Under $H_G^{r/K}$ at least m cohorts are null. In those, $z_k \sim N(0,1)$ exactly (by Proposition 1, applied with $\Phi^{-1} \circ F_t$), so their one-sided p-values are independent and uniform, given the external structure. By pigeonhole, the j-th largest of all K p-values is at least the j-th largest of the null ones, for $j = 1, \ldots, m$. The m largest of $m_0 \geq m$ independent uniforms are stochastically no smaller than m sorted independent uniforms. The Stouffer p-value is non-decreasing in each argument, so it is super-uniform. The factor 2 is a Bonferroni correction over s (Benjamini and Heller, 2008). The argument in fact holds pointwise: for any m null cohorts, the j-th largest of all K p-values is at least the j-th largest within that subset, and m independent uniforms give an exactly $N(0,1)$ Stouffer sum. If G is untested in some cohorts, condition on the set of tested cohorts. Each z_k is exactly $N(0,1)$ conditionally on (X_{-G}, Y, b) of its cohort, so a selection rule that is a function of those variables leaves the tested null z_k independent $N(0,1)$, and the argument applies with K replaced by the number of tested cohorts. A rule that uses X_G is not covered.

Proof sketch of Proposition 2. Write $X_G = D_G\beta + U\gamma + \mathcal{E}$ with $g = 1$. Then

$$u = \tilde{\eta}^\top(\tilde{U}\gamma + \tilde{\mathcal{E}})/\|\tilde{\eta}\|.$$

The term $\tilde{\eta}^\top\tilde{\mathcal{E}}/\|\tilde{\eta}\| \sim N(0, \sigma_c^2)$ as before. The term $\tilde{\eta}^\top\tilde{U}\gamma/\|\tilde{\eta}\| = \sqrt{n}\,\gamma\,\hat{\kappa}$, with $\hat{\kappa} = \widehat{\mathrm{Cov}}(\eta, U \mid D)/\widehat{\mathrm{sd}}(\eta \mid D) \to \kappa$. The residual sum $S = \tilde{X}_G^\top Q \tilde{X}_G$ contains U residualised on (D_G, η), so $S/\nu \to \sigma_c^2 + \gamma^2\tau^2$. Hence $t = \sqrt{\nu}\,u/\sqrt{S} = \sqrt{n}\,\gamma\kappa/\sqrt{\sigma_c^2 + \gamma^2\tau^2} + O_p(1)$, using $\nu/n \to 1$. Strictly, t is a doubly non-central t variable; the statement only describes its centring.

B Additional material

Full simulation grids, the knockoff negative result (G0), the duplicate-screen calibration, per-cohort permutation rates and the protocol are in the repository (`results/`, `docs/`).